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United States Patent	12416353
Kind Code	B2
Date of Patent	September 16, 2025
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Case

Abstract

A case that houses a drive force transmission device includes a circular ring part, fastening boss parts, first ribs, a non-fastening boss part, and a second rib. The circular ring part defines an opening through which a shaft is configured to be inserted. The fastening boss parts are provided around the circular ring part. Fastening members are to be secured to the fastening boss parts. The first ribs are respectively formed between a portion of the fastening boss parts and the circular ring part when viewed from an axial direction of the shaft. The non-fastening boss part is provided between adjacent ones of the fastening boss parts to which the first ribs are not connected, when viewed from the axial direction. A fastening member is not configured to be secured to the non-fastening boss part. The second rib is formed between the non-fastening boss part and the circular ring part.

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Appl. No.:	18/291993
Filed (or PCT Filed):	June 22, 2022
PCT No.:	PCT/JP2022/024983
PCT Pub. No.:	WO2023/037695
PCT Pub. Date:	March 16, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240369131 A1	Nov. 07, 2024

Foreign Application Priority Data

JP 2021-146523 Sep. 08, 2021

Publication Classification

Int. Cl.: F16H57/02 (20120101); F16H57/03 (20120101)

U.S. Cl.:

CPC F16H57/02 (20130101); F16H57/03 (20130101); F16H2057/02017 (20130101)

Field of Classification Search

CPC: F16H (57/02); F16H (57/03); F16H (2057/02017); F16H (2057/02021)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3246538	12/1965	Easton	74/606R	F16H 1/20
4108021	12/1977	MacAfee	74/606R	B60K 17/3467
8911312	12/2013	Ito	474/93	F16H 57/0489

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2000-110922	12/1999	JP	N/A
2005-172125	12/2004	JP	N/A
2017-082958	12/2016	JP	N/A

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Background/Summary

(1) This is a U.S. national phase application of PCT/JP2022/024983, filed on Jun. 22, 2022, which claims priority to Japanese Patent Application No. 2021-146523, filed on Sep. 8, 2021. The entire disclosure of Japanese Patent Application No. 2021-146523 is hereby incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a case.

BACKGROUND ART

(3) Japanese Laid-Open Patent Publication No. 2017-82958 discloses a transmission case having an opening through which a drive shaft is inserted. This transmission case has a rib that extends radially from a circular ring part that forms the opening, and connects to a bolt boss for fastening.

SUMMARY

- (4) This case has ribs provided between a circular ring part that forms an opening through which a shaft is inserted, and bolt bosses for fastening, increasing the strength. In this case, if holes for a component to pass through or thick parts for an oil passage to pass through are formed between the bolt boss and the circular ring part, locations may occur where ribs cannot be provided.
- (5) When manufacturing this kind of case by casting, molten metal inside a mold is supplied to each part via recesses corresponding to the ribs. Molten metal is not easily supplied to locations where there are no recesses corresponding to the ribs. As a result, imbalance occurs in the flow of the molten metal inside the mold, and there is a risk of cavities occurring in the case after molding. If cavities occur, the yield of cases decreases.
- (6) In light of that, there is a demand to provide a case that reduces the occurrence of cavities and improves yield.
- (7) The case of an embodiment of the present disclosure is a case that houses a drive force transmission device, including: a circular ring part that forms an opening through which a shaft that transmits a drive force is inserted, a plurality of fastening boss parts provided around the circular ring part, to which fastening members are secured, first ribs that, when viewed from the axial direction of the shaft, are formed between a portion of the fastening boss parts among the plurality of fastening boss parts and the circular ring part, a non-fastening boss part that, when viewed from the axial direction, is provided between the plurality of other fastening boss parts among the plurality of fastening boss parts to which the first ribs are not connected, and to which the fastening member is not secured, and a second rib that is formed between the non-fastening boss part and the circular ring part.
- (8) According to an embodiment of the present disclosure, it is possible to provide a case that reduces the occurrence of cavities and improves yield.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a drawing for explaining an automatic transmission case.
- (2) FIG. 2 is a drawing for explaining a transmission case.
- (3) FIG. 3 is a drawing for explaining the transmission case.
- (4) FIG. 4 is a drawing for explaining the transmission case.
- (5) FIG. 5 is a drawing for explaining the transmission case.
- (6) FIG. 6 is an enlarged view of the main parts of the transmission case.
- (7) FIG. 7 is a drawing for explaining the ribs.
- (8) FIG. 8 is a drawing for explaining a mold.
- (9) FIG. 9 is a drawing for explaining the mold.
- (10) FIG. 10 is a drawing for explaining the flow of molten metal inside the mold.
- (11) FIG. 11 is a drawing for explaining the flow of molten metal inside the mold.
- (12) FIG. 12 is a drawing for explaining the flow of molten metal inside the mold.
- (13) FIG. 13 is a drawing for explaining the flow of molten metal inside the mold.
- (14) FIG. 14 is a drawing for explaining the transmission case according to a comparison example.
- (15) FIG. 15 is a drawing for explain the flow of molten metal inside the mold according to a comparison example.
- (16) FIG. 16 is a drawing for explain the flow of molten metal inside the mold according to a comparison example.
- (17) FIG. 17 is a drawing for explain the flow of molten metal inside the mold according to a comparison example.
- (18) FIG. 18 is a drawing for explaining the transmission case according to a modification example.

DETAILED DESCRIPTION OF EMBODIMENTS

(19) Hereafter, the present embodiment is explained.

(20) FIG. 1 is a drawing for explaining an automatic transmission case 4.

(21) FIG. 2 is a drawing for explaining a transmission case 1. In FIG. 2, the transmission case 1 is shown in cross section, and a transmission 61, an output shaft 63, a control valve CV, and an oil pan 5 are shown using virtual lines.

(22) FIG. 3 is a drawing for explaining the transmission case 1. FIG. 3 is the A-A arrow view of FIG. 2. In FIG. 3, cross hatching is applied to a lower end surface 13a of a wall part 13.

(23) FIG. 4 is a drawing for explaining the transmission case 1. FIG. 4 is the B-B arrow view of FIG. 2. In FIG. 4, cross hatching is applied to a support part 155, a junction part 12, and ribs 7, 8.

(24) FIG. 5 is a drawing for explaining the transmission case 1. FIG. 5 is a perspective view of the transmission case 1 seen from a cover member 3 side.

(25) FIG. 6 is a drawing for explaining the transmission case 1. FIG. 6 is an enlarged view of region A of FIG. 4. In FIG. 6, to make it easier to understand the positional relationship, cross hatching is applied to the ribs 7, 8, the support part 155, and an outer diameter side wall part 151.

(26) FIG. 7 is a drawing for explaining the transmission case 1. FIG. 7 is a schematic diagram of the A-A cross section of FIG. 6. In FIG. 7, cross hatching is applied to the rib 7.

(27) As shown in FIG. 1, the automatic transmission case 4 accommodates a torque converter 60, an input shaft 62, the transmission 61, and the output shaft 63 (shaft) which are a drive force transmission device 6 (see dashed line in FIG. 1).

(28) In the drive force transmission device 6, the torque converter 60, the input shaft 62, the transmission 61, and the output shaft 63 are provided along the output rotation transmission route around a rotation axis X of an engine (not illustrated).

(29) In the drive force transmission device 6, the output rotation of the engine is transmitted to the transmission 61 via the input shaft 62 from the torque converter 60. The output rotation transmitted from the transmission 61 is shifted by changing the combination of engagement/release of a plurality of friction engagement elements (not illustrated) constituting the transmission 61.

(30) The output rotation shifted by the transmission 61 is transmitted to the drive wheels of a vehicle (not illustrated) via the output shaft 63.

(31) As shown in FIG. 1, the automatic transmission case 4 has a converter housing 2 that accommodates the torque converter 60, the transmission case 1 that accommodates the transmission 61, and the cover member 3 that accommodates the output shaft 63.

(32) With the installation state of the automatic transmission case 4 in the vehicle as a reference, the oil pan 5 that stores lubricating oil (not illustrated) is secured to the bottom of the transmission case 1 in a vertical line VL direction (in the drawing, the up-down direction).

(33) The transmission case 1 has a cylinder wall part 10 surrounding the rotation axis X.

(34) The cylinder wall part 10 has junction parts 11, 12 respectively on one end 10a side and another end 10b side in the rotation axis X direction.

(35) A junction part 20 of the converter housing 2 abuts the junction part 11 from the rotation axis X direction. These junction parts 11, 20 are linked to each other by a bolt B which is a fastening member.

(36) As shown in the enlarged area in FIG. 1, a bolt boss 121 is provided on the junction part 12. A bolt hole 120 of the bolt boss 121 is opened at the other end 10b of the cylinder wall part 10. The cross section around the bolt boss 121 is shown in the enlarged area in FIG. 1.

(37) A plurality of the bolt bosses 121 and the bolt holes 120 are formed in the circumferential direction around the rotation axis X (see FIG. 4).

(38) As shown in FIG. 1, a junction part 30 of the cover member 3 abuts the other end 10b of the cylinder wall part 10 from the rotation axis X direction. The cover member 3 has a cylindrical shape. The junction part 30 is a flange surrounding the outer circumference of the cover member 3.

(39) As shown in FIG. 1, through holes 31 that penetrate in the rotation axis X direction are formed

on the junction part **30** of the cover member **3**. The transmission case **1** and the cover member **3** are secured to each other by screwing in the bolts **B** from the cover member **3** side in a state with the center lines of the through holes **31** of the junction part **30** matched to center lines **C** of the bolt holes **120** of the junction part **12**.

(40) As shown in FIG. 2, a wall part **15** is provided extending to the inner diameter side on the other end **10b** side of the cylinder wall part **10**. The wall part **15** is provided in a direction orthogonal to the rotation axis **X**.

(41) The cylinder wall part **10** accommodates the transmission **61** in the interior space on the one end **10a** side from the wall part **15** in the rotation axis **X** direction (see virtual lines in the drawing). In the cylinder wall part **10**, the region on the other end **10b** side from the wall part **15** in the rotation axis **X** direction serves as the junction part **12** with the cover member **3** (see FIG. 1).

(42) An opening **150** is formed in a region intersecting with the rotation axis **X** of the wall part **15**. The output shaft **63** (see virtual lines in the drawing) is inserted through the opening **150**. The output shaft **63** crosses the wall part **15** from the one end **10a** side of the cylinder wall part **10** to the other end **10b** side.

(43) The support part **155** that surrounds the opening **150** is provided on the surface of the wall part **15** on the other end **10b** side (right side in the drawing) of the cylinder wall part **10**. The support part **155** protrudes in a direction away from the wall part **15** in the rotation axis **X** direction. As shown in FIG. 4, the support part **155** has a circular ring shape when viewed from the rotation axis **X** direction. Specifically, the support part **155** constitutes a circular ring part.

(44) As shown in FIG. 2, a bearing **B1** is provided on the inner circumference of the support part **155**. The outer circumference of the output shaft **63** is supported by the support part **155** via the bearing **B1**.

(45) The wall part **15** bulges in a direction gradually approaching the other end **10b** side of the cylinder wall part **10** in the rotation axis **X** direction as it goes from the outer diameter side of the rotation axis **X** toward the inner diameter side.

(46) In specific terms, the wall part **15** is provided with the outer diameter side wall part **151** connected to the cylinder wall part **10** at the radial outer diameter side of the rotation axis **X**, an inner diameter side wall part **153** connected to the support part **155** on the radial inner diameter side of the rotation axis **X**, and a connection wall part **152** that connects the outer diameter side wall part **151** and the inner diameter side wall part **153**.

(47) In the rotation axis **X** direction, the inner diameter side wall part **153** is offset more to the other end **10b** side of the cylinder wall part **10** in the rotation axis **X** direction than the outer diameter side wall part **151**. The connection wall part **152** is positioned between the outer diameter side wall part **151** and the inner diameter side wall part **153** in the rotation axis **X** direction. The connection wall part **152** is connected to the outer diameter side wall part **151** and the inner diameter side wall part **153** respectively via step parts **D**, **D**.

(48) In the present embodiment, the outer diameter side wall part **151**, the connection wall part **152**, and the inner diameter side wall part **153** have the thickness of the portion extending along the radial direction of the rotation axis **X** set to **T1**. The thickness **T1** is the thickness in the rotation axis **X** direction. The thickness **T1** of the outer diameter side wall part **151**, the connection wall part **152**, and the inner diameter side wall part **153** may be different from each other.

(49) As shown in FIG. 2, the wall part **13** is provided on the bottom of the cylinder wall part **10** in the vertical line **VL** direction. The wall part **13** is formed integrally with the cylinder wall part **10**. The wall part **13** extends in the direction away from the cylinder wall part **10** along the vertical line **VL** direction.

(50) As shown in FIG. 3, when the transmission case **1** is viewed from below in the vertical line **VL** direction, the wall part **13** has substantially a rectangular shape. The wall part **13** is configured from long wall parts **131**, **132** provided facing along the rotation axis **X** and short wall parts **133**, **134** that connect the end parts of these long wall parts **131**, **132** to each other. The region surrounded by

the long wall parts **131**, **132** and the short wall parts **133**, **134** constitutes the bottom opening of the transmission case **1**.

(51) As shown in FIG. 2, a communication hole **105** that communicates between the interior and exterior of the cylinder wall part **10** is formed in the region surrounded by the wall part **13** in the cylinder wall part **10**.

(52) As shown in FIG. 3, when viewed from below the cylinder wall part **10**, the communication hole **105** is formed inside the region surrounded by the wall part **13**.

(53) As shown in FIG. 2, the oil pan **5** is joined to the lower end surface **13a** of the wall part **13**. The control valve CV is accommodated in the space surrounded by the wall part **13** and the oil pan **5** (see virtual lines in FIG. 2).

(54) As shown in FIG. 3, the short wall part **134** has a width **W1** that is substantially consistent with a length **L1** of the junction part **12** (see FIG. 2) in the rotation axis X direction. A width **W2** of the long wall parts **131**, **132** and the short wall part **133** is narrower than the width **W1** of the short wall part **134** ($W2 < W1$). An outer wall surface **134a** of the short wall part **134** is orthogonal to the rotation axis X, and is flush with the other end **10b** of the cylinder wall part **10** (see FIG. 2, FIG. 5).

(55) As shown in FIG. 3 and FIG. 5, three oil passages Ha, Hb, Hc are opened on the lower end surface **13a** on the short wall part **134** side of the wall part **13**. The oil passages Ha, Hb, Hc are aligned along the direction intersecting with the rotation axis X.

(56) As shown in FIG. 2, the oil passage Hc is provided facing along the vertical line VL direction inside the wall part **15**. The lower end in the vertical line VL direction is in communication with an oil passage (not illustrated) provided in the control valve CV. The top end of the oil passage Hc is in communication with a case-internal oil passage (not illustrated) that passes through the inside of the transmission case **1**. In FIG. 2, the oil passage Hc is illustrated, but oil passages Ha, Hb are also the same.

(57) As shown in FIG. 3, thick parts **135**, **136**, and **137** for which the thickness in the rotation axis X direction has been increased are provided in the region through which the oil passages Ha, Hb, Hc pass in the wall part **15**. A thickness **T2** in the thick parts **135**, **136**, **137** in the rotation axis X direction is thicker than the thickness **T1** of the wall part **15** ($T2 > T1$). The thick parts **135**, **136**, **137** bulge from the surfaces of one side and the other side of the wall part **15** in the rotation axis X direction.

(58) As shown in FIG. 4, the thick parts **135**, **136**, **137** bulge to the paper surface front side from the wall part **15** in the region below a horizontal line HL through which the rotation axis X passes.

(59) A thick part **138** is provided at a position adjacent to the thick part **137** in the wall part **15** in the circumferential direction around the rotation axis X. The thick part **138** also bulges to the paper surface front side from the wall part **15**.

(60) The thick part **138** extends in the rotation axis X radial direction. An oil passage (not illustrated) passes through the interior of the thick part **138**. The oil passage inside the thick part **138** communicates between the opening **150** and the accommodation space of the control valve CV (see FIG. 2).

(61) Furthermore, a through hole **139** is provided at a position adjacent to the thick part **138** in the wall part **15** in the circumferential direction around the rotation axis X. The through hole **139** penetrates the wall part **15** in the rotation axis X direction. A harness of electrical equipment, etc., (not illustrated) is inserted through the through hole **139**.

(62) Here, as shown in FIG. 4, when the transmission case **1** is viewed from the cover member **3** side, the short wall part **134** extends facing along a straight line HL1 parallel to the horizontal line HL.

(63) As shown in FIG. 4 and FIG. 5, the junction part **12** of the transmission case **1** is connected to the short wall part **134**. In specific terms, as shown in FIG. 4, the junction part **12** has parallel walls **127**, **128** extending along straight lines VL1, VL2 below the horizontal line HL. The straight lines VL1, VL2 are straight lines parallel to the vertical line VL that passes through the rotation axis X.

The lower ends of the parallel walls **127**, **128** are respectively connected to one end part **134b** and another end part **134c** of the short wall part **134** in the straight line HL1 direction.

(64) The junction part **12** has an arc-shaped wall **129** that surrounds the support part **155** above the horizontal line HL that passes through the rotation axis X. Both ends of the arc-shaped wall **129** in the circumferential direction around the rotation axis X are connected to the top ends of the parallel walls **127**, **128**. Viewed from the rotation axis X direction, the junction part **12** (arc-shaped wall **129**, parallel walls **127**, **128**) and the short wall part **134** form one continuous wall surrounding the support part **155**.

(65) As shown in FIG. 4, the plurality of bolt bosses **121** are provided on the junction part **12**. The plurality of bolt bosses **121** are provided on the junction part **12** at intervals in the circumferential direction around the rotation axis X. As shown in FIG. 6, viewed from the rotation axis X direction, the bolt bosses **121** bulge to the radial outer diameter side of the rotation axis X from an outer peripheral surface **12a** of the junction part **12**. The bolt bosses **121** bulge smoothly from the outer peripheral surface **12a** via an arc-shaped recess **121a**. Viewed from the rotation axis X direction, the curvature radius of the recess **121a** is $r2$.

(66) As shown in FIG. 4, a bolt boss **122a** is provided on the one end part **134b** of the short wall part **134** in the straight line HL1 direction. A bolt boss **122c** is provided on the other end part **134c** of the short wall part **134**. A bolt boss **122b** is provided between the one end part **134b** and the other end part **134c** of the short wall part **134**.

(67) The parallel wall **127** of the junction part **12** and the one end part **134b** of the short wall part **134** are connected via the bolt boss **122a**. The parallel wall **128** of the junction part **12** and the other end part **134c** of the short wall part **134** are connected via the bolt boss **122c**.

(68) As shown in FIG. 6, an arc-shaped recess **122d** recessed to the radial inner diameter side is formed on the outer peripheral surface of the portion at which the bolt boss **122c** and the short wall part **134** are continuous. Viewed from the rotation axis X direction, a curvature radius $r1$ of the recess **122d** is larger than the curvature radius $r2$ of the recess **121a** ($r1 > r2$).

(69) As shown in FIG. 4, the bolt holes **120** having a common hole diameter are formed on the plurality of bolt bosses **121** and three bolt bosses **122a**, **122b**, and **122c**. The cover member **3** is secured to the bolt bosses **122a** to **122c** by screwing in the bolts B (see FIG. 1). Specifically, the plurality of bolt bosses **121** and the three bolt bosses **122a** to **122c** constitute a plurality of fastening boss parts to which the bolts B, which are fastening members, are secured.

(70) A pin boss **123** is provided on the arc-shaped wall **129** of the junction part **12** at one location between bolt bosses **121**, **121** adjacent in the circumferential direction. A pin hole **123a** (recess) is formed on the pin boss **123**. A pin P1 for positioning is inserted in the pin hole **123a**.

(71) As shown in FIG. 4, a pin boss **124** is provided on the other end part **134c** side in the short wall part **134**. As shown in FIG. 6, a pin hole **124a** (recess) is formed in the pin boss **124**. A pin P2 for positioning is inserted in the pin hole **124a**.

(72) In FIG. 4 and FIG. 6, hatching is applied to pins P1, P2. Examples of the pins P1, P2 include well-known positioning pins such as knock pins, parallel pins, etc. The diameter of the pins P1, P2 may be the same or may be different. Specifically, the pin bosses **123**, **124** constitute a non-fastening boss part to which the bolts B which are fastening members are not secured.

(73) As shown in FIG. 4, the pin boss **124** is provided at a position offset by a distance CL1 in the straight line HL1 direction to the bolt boss **122b** side from the bolt boss **122c**. The pin boss **124** is continuous with the bolt boss **122c**.

(74) The pin bosses **123**, **124** through which the pins P1, P2 are inserted are disposed at symmetrical positions across the vertical line VL and the horizontal line HL. The transmission case **1** and the cover member **3** are secured to each other using the bolts B (see FIG. 1) after being positioned in the circumferential direction around the rotation axis X by the pins P1, P2. The transmission case **1** has ribs **7**, **8**, **9** for increasing the rigidity of the fixed part with the cover member **3**.

(75) [Rib 7 (First Rib)]

(76) As shown in FIG. 4, a plurality of the ribs 7 (first ribs) are provided between the junction part 12 and the support part 155 in the radial direction of the rotation axis X. Each of the ribs 7 is provided facing along the radial direction of the rotation axis X. The ribs 7 are connected to the support part 155 on the radial inner diameter side of the rotation axis X, and are connected to the bolt bosses 121 of the junction part 12 at the radial outer diameter side of the rotation axis X. Specifically, the bolt bosses 121 constitute the portion of the fastening boss parts to which the first ribs are connected among the plurality of fastening boss parts.

(77) As shown in FIG. 2, the ribs 7 protrude in the rotation axis X direction from the surface on the junction part 12 side of the wall part 15. The ribs 7 are provided straddling the junction part 12, the wall part 15, and the support part 155 in the radial direction of the rotation axis X.

(78) Here, as shown in FIG. 4, the through hole 139 is provided between the support part 155 and the bolt boss 122a in the radial direction of the rotation axis X. The thick part 138 is provided between the support part 155 and the bolt boss 122b in the radial direction of the rotation axis X. Therefore, the ribs 7 are not connected to the bolt bosses 122a, 122b, respectively.

(79) The rib 9 described later is provided between the support part 155 and the bolt boss 122c in the radial direction of the rotation axis X. The rib 9 is a rib that differs from the ribs 7. Specifically, the ribs 7 which are the first ribs are not connected to the bolt bosses 122a to 122c of the short wall part 134. Therefore, the bolt bosses 122a to 122c of the short wall part 134 constitute the plurality of other fastening boss parts to which the first ribs are not connected among the plurality of fastening boss parts.

(80) [Rib 8 (Second Rib)]

(81) As shown in FIG. 4, the rib 8 (second rib) is provided between the short wall part 134 and the support part 155 in the radial direction of the rotation axis X. The rib 8 is provided along the radial direction of the rotation axis X. As shown in FIG. 6, a width T8 of the rib 8 substantially matches a width T7 of the rib 7 ($T8 \approx T7$).

(82) The rib 8 is connected to the support part 155 at the radial inner diameter side of the rotation axis X, and is connected to the pin boss 124 at the radial outer diameter side of the rotation axis X.

(83) As shown in FIG. 4, the pin boss 124 is provided between bolt bosses 122b, 122c. Specifically, the pin boss 124 is provided between the plurality of other boss parts to which the ribs 7 are not connected.

(84) As shown in FIG. 4, the rib 8 has a branched rib 85 branched from between the support part 155 and the pin boss 124 in the radial direction of the rotation axis X. The branched rib 85 extends in the straight line HL1 direction and is connected to the thick part 136.

(85) As shown in FIG. 2, the rib 8 protrudes in the rotation axis X direction from the surface on the junction part 12 side of the wall part 15. The rib 8 is provided straddling the short wall part 134, the wall part 15, and the support part 155 in the radial direction of the rotation axis X.

(86) [Rib 9 (Third Rib)]

(87) As shown in FIG. 4, the rib 9 (third rib) is provided between the rib 7 and the rib 8 adjacent in the circumferential direction around the rotation axis X.

(88) As shown in FIG. 6, the rib 9 has a radial direction wall 91 extending in the radial direction of the rotation axis X, and a circumferential direction wall 92 extending in the circumferential direction around the rotation axis X. The radial direction wall 91 connects the bolt boss 122c and the step part D between the outer diameter side wall part 151 and the connection wall part 152 in the wall part 15. The circumferential direction wall 92 connects the bolt boss 121 to which the rib 7 is connected and the radial direction wall 91.

(89) As shown in FIG. 6, the radial direction wall 91 and the circumferential direction wall 92 have a substantially same width T9.

(90) The width T9 of the rib 9 (radial direction wall 91, circumferential direction wall 92) is thinner than widths T7, T8 of ribs 7, 8 ($T9 < T7$, $T9 < T8$).

(91) An intersection part K of the radial direction wall **91** and the circumferential direction wall **92** is positioned substantially in the middle of the bolt bosses **121**, **122c** in the vertical line VL direction. The rib **9** constitutes a truss structure that connects three points of the intersection part K, the bolt boss **122c**, and the bolt boss **121**.

(92) As shown in FIG. 6, a straight line Lp1 along the longitudinal direction of the radial direction wall **91** connects to the bolt boss **122c** by passing above a center C1 of the bolt hole **120** of the bolt boss **122c** in the vertical line VL direction. A straight line Lp2 along the longitudinal direction of the circumferential direction wall **92** connects to the bolt boss **121** by passing below a center C2 of the bolt hole **120** of the bolt boss **121** in the vertical line VL direction **1**.

(93) Viewed from the rotation axis X direction, the radial direction wall **91** and the circumferential direction wall **92** are provided at positions respectively offset approaching each other in the vertical line VL direction from the center C1 of the bolt boss **122c** and the center C2 of the bolt boss **121**.

(94) As shown in FIG. 6, the abovementioned thick part **135** is provided in the region in which the rib **9** is provided in the wall part **15**. The thick part **135** bulges to the paper surface front side from the outer diameter side wall part **151** of the wall part **15**. The circumferential direction wall **92** of the rib **9** is provided in the region avoiding the thick part **135** in the outer diameter side wall part **151**.

(95) As shown in FIG. 7, the outer diameter side wall part **151** has a bottom surface **151a** orthogonal to the rotation axis X on the rib **7** side from the center C2 of the bolt boss **121** in the radial direction of the rotation axis. The outer diameter side wall part **151** has an inclined plane **151b** inclined in a direction approaching the other end **10b** in the rotation axis X direction on the parallel wall **128** side from the center C2 of the bolt boss **121** in the radial direction of the rotation axis X.

(96) The inclined plane **151b** is inclined in a direction approaching the other end **10b** in the rotation axis X direction as it moves away from the bottom surface **151a** in the radial direction of the rotation axis X.

(97) For example, when the circumferential direction wall **92** is provided at a position overlapping the center C2, the circumferential direction wall **92** is disposed on the bottom surface **151a**. In contrast to this, in the present embodiment, the circumferential direction wall **92** is provided on the inclined plane **151b** offset in the radial direction of the rotation axis X from the center C2. As a result, the protrusion height of the circumferential direction wall **92** from the outer diameter side wall part **151** in the rotation axis X direction becomes shorter by ΔH . As a result, the amount of material used for forming the circumferential direction wall **92** is reduced, so it can be made lighter.

(98) Hereafter, casting of the transmission case **1** is explained.

(99) FIG. 8 is a drawing for explaining a mold used for casting the transmission case **1**.

(100) FIG. 9 is a drawing for explaining the mold used for casting the transmission case **1**. FIG. 9 is the A-A arrow view of FIG. 8.

(101) FIG. 10 is a drawing for explaining the flow of a molten metal Q inside a mold M.

(102) FIG. 11 is a drawing for explaining the flow of the molten metal Q inside the mold M. FIG. 11 is the A-A arrow view of FIG. 10.

(103) FIG. 12 and FIG. 13 are drawings for explaining the flow of the molten metal Q inside the mold M in sequence.

(104) In FIG. 8 to FIG. 13, the interior space of the mold M corresponding to the shape of the transmission case **1** is shown by dashed lines. In FIG. 10 to FIG. 13, hatching is applied to the region inside the mold M to which the molten metal Q is supplied.

(105) The transmission case **1** is manufactured by casting by flowing the molten metal Q into the mold M. The molten metal Q component is a magnesium alloy.

(106) The mold M has an interior space corresponding to the shape of the transmission case **1** (transmission case **1'**, see dashed lines in FIG. 8). In the explanation hereafter, “'” is added to the notation of the code of the part corresponding to each part of the transmission case **1** in the interior

space of the mold M.

(107) As shown in FIG. 8, the mold M is disposed with the rotation axis X along the vertical line VL direction.

(108) In specific terms, in the transmission case '1, another end 10b' side of a cylinder wall part 10' is disposed on the upper side in the vertical line VL direction, and one end side 10a' side is disposed on the lower side of the vertical line VL direction.

(109) In the transmission case 1', a lower end surface 13a' of a wall part 13' is disposed at one end face side (right side in the drawing) in the radial direction of the rotation axis X.

(110) As shown in FIG. 8 and FIG. 9, entry ports M1, M2, M3 for the molten metal Q are provided at three locations and outlet port M4 for the molten metal Q is provided at one location in the mold M. The entry ports M1, M2, M3 and the outlet port M4 are through holes that communicate between the interior and exterior of the mold M.

(111) As shown in FIG. 8, the entry port M1 is connected to the wall part 13' at the lower side (one end 10a' side) of the mold M in the vertical line VL direction. The entry port M1 is provided in a direction along the rotation axis X.

(112) As shown in FIG. 8, the entry port M2 is connected to the lower end surface 13a' of the wall part 13' at the upper side (other end 10b' side) of the mold M in the vertical line VL direction. The entry port M2 is provided in a direction along the straight line Lm that crosses the rotation axis X from the one side (wall part 13') to the other side (cylinder wall part 10').

(113) As shown in FIG. 9, the entry port M2 is connected to a one end part 134b' of a short wall part 134'.

(114) As shown in FIG. 9, the entry port M3 is connected to another end part 134c' of the short wall part 134'. The entry port M3, the same as entry port M2, is provided in a direction along the straight line Lm that crosses the rotation axis X from one side to the other side. The entry port M3 is provided at the same position as the entry port M2 (see FIG. 8) in the vertical line VL direction, and is connected to the lower end surface 13a'.

(115) As shown in FIG. 8, the outlet port M4 is provided at the opposite side to the wall part 13' across the rotation axis X at the upper side (other end 10b' side) of the mold M in the vertical line VL direction. The outlet port M4 is provided in a direction along the straight line Lm.

(116) In specific terms, as shown in FIG. 9, the outlet port M4 is connected to the intersection point of an arc-shaped wall 129' of a junction part 12' and the straight line Lm. As shown in FIG. 8 and FIG. 9, a case when the mold M has the entry ports M1, M2, M3 for the molten metal Q at three locations was explained as embodiments of the present invention, but the present invention is not limited only to a mode showing these embodiments, and it is also possible to further provide one or a plurality of entry ports between the entry port M1 and the entry port M2, or between the entry port M1 and the entry port M3 according to the requirements for casting.

(117) The effects of the transmission case 1 of this configuration are explained.

(118) As shown in FIG. 10 and FIG. 11, the molten metal Q is injected from entry ports M1 to M3 in the space inside the mold M (transmission case 1'). Most of the molten metal Q first flows in a direction along the injection direction of the molten metal Q inside the transmission case 1'.

(119) As shown in FIG. 10, most of the molten metal Q injected from the entry port M1 flows along the wall part 13' in the rotation axis X direction (arrow A). As a result, around the wall part 13' is filled with the molten metal Q. Thereafter, the molten metal Q flows to the cylinder wall part 10' side (arrow B) from the wall part 13' side along the radial direction of the rotation axis X.

(120) As shown in FIG. 10, the molten metal Q injected from the entry port M2 flows to the cylinder wall part 10' side from the wall part 13' side (arrow C) along the radial direction of the rotation axis X. The molten metal Q injected from the entry port M3 similarly flows in the arrow C direction (see FIG. 11).

(121) As shown in FIG. 11, immediately after being supplied from entry ports M2, M3, the molten metal Q is first filled in the short wall part 134'. Thereafter, it flows from the short wall part 134' to

the junction part **12'**, a wall part **15'**, and ribs **7'**, **8'**.

(122) Here, with the transmission case **1**, the thickness (**W1**, **L1**) of the short wall part **134**, the junction part **12**, and the ribs **7**, **8** in the rotation axis X direction is thicker than the thickness **T1** of the wall part **15** (see FIG. 2, FIG. 3). Specifically, in the space inside the mold M (transmission case **1'**), the short wall part **134'**, the junction part **12'**, and the ribs **7'**, **8'** have a larger cross section area as a flow path through which the molten metal Q flows than the wall part **15'**.

(123) Therefore, the molten metal Q that flows from the short wall part **134'** passes more easily through parallel walls **127'**, **128'** of the junction part **12'**, and the ribs **7'**, **8'** which have a large flow path cross section area (arrows C, D in FIG. 11).

(124) Meanwhile, since there are thick parts **137'**, **138'**, items correlating to ribs **7'**, **8'** are not provided on a bolt boss **122b'**. The bolt boss **122b'** is at a position away from the entry ports **M2**, **M3**.

(125) Therefore, though the region around the bolt boss **122b'** in the wall part **15'** is supplied with the molten metal Q from the short wall part **134'**, the supply rate of the molten metal Q is slower than with regions provided with the ribs **7'**, **8'**. That being done, as shown in FIG. 11, inside the transmission case **1'**, the molten metal Q is in a state in which the region around the bolt boss **122b'** in the wall part **15'** is recessed in the direction away from the outlet port **M4** in the straight line Lm direction.

(126) Here, a gas G is generated from the molten metal Q. The gas G is pushed out by the molten metal Q sequentially supplied into the mold M, and exhausted from the outlet port **M4** (see FIG. 12).

(127) A detailed description will be noted later, but when a difference (imbalance) occurs in the supply rate of the molten metal Q to each part inside the mold M, the gas G is not pushed out appropriately, and casting may be done with the gas G still contained (see FIG. 17). The caught gas G appears as cavities in the transmission case after casting. The cavities occur more easily the greater the difference in the supply rate of the molten metal Q.

(128) As shown in FIG. 4, with the transmission case **1** of the present embodiment, the rib **8** is connected to the pin boss **124**. The pin boss **124** is offset by the distance **CL1** to the bolt boss **122b** side from the bolt boss **122c**.

(129) As shown in FIG. 11, inside the mold M, the rib **8'** is disposed closer to the bolt boss **122b'** than when connected to a bolt boss **122c'**. As a result, it is easier to supply a portion of the molten metal Q that passes through the rib **8'** to the region around the bolt boss **122b'** than when the rib **8'** is connected to the bolt boss **122c'**. A branched rib **85'** is connected to the rib **8'**, so supply to the region around the bolt boss **122b'** is promoted more.

(130) The bolt boss **122c** and the pin boss **124** are formed continuously (see FIG. 4). As a result, the space around the boss **124'** (capacity) inside the mold M is greater than when the bolt boss **122c** and the pin boss **124** are provided separately, for example.

(131) Having done that, the amount of the molten metal Q supplied to the rib **8'** from the pin boss **124'** increases, so supply of the molten metal Q to the region around the bolt boss **122b'** in the wall part **15'** is promoted.

(132) In this way, by having the transmission case **1** have the rib **8** provided on the pin boss **124**, during casting, it is possible to actively supply the molten metal Q also to the region around the bolt boss **122b'** in the wall part **15'** in which it is difficult to supply the molten metal Q. As a result, there is a reduction in the difference in the supply rate of the molten metal Q to each part within the mold M becoming large, and it is possible to sequentially push out the gas G generated from the molten metal Q toward the outlet port **M4** side (see FIG. 12). As a result, there is a reduction in the occurrence of cavities in the transmission case **1** after casting (see FIG. 13).

COMPARISON EXAMPLE

(133) FIG. 14 is a drawing for explaining a transmission case **100** according to a comparison example.

(134) FIG. 15 to FIG. 17 are drawings for explaining the casting of the transmission case **100** in sequence according to the comparison example. In the comparison example hereafter, only the parts that are different from the present embodiment are explained.

(135) As shown in FIG. 14, a rib **800** of the transmission case **100** is connected to the bolt boss **122c**. The rib **800** is provided at a position farther away from the bolt boss **122b** than the pin boss **124** in the straight line HL1 direction by the distance CL1 for the part connected to the bolt boss **122c**.

(136) As shown in FIG. 15 to FIG. 17, the transmission case **100** is formed by flowing the molten metal Q into a mold MA. The mold MA has a space corresponding to the shape of the transmission case **100** (transmission case **100'**, see dashed lines in FIG. 15 to FIG. 17).

(137) Inside the mold MA, in the transmission case **100'**, the supply to the region around the bolt boss **122b'** in the wall part **15'** is delayed by the amount that a rib **800'** is separated from the bolt boss **122b'**.

(138) The molten metal Q supplied to the rib **800'** from the inlet port M3 reaches a support part **155'** (see arrow D in FIG. 15) before reaching the region around the bolt boss **122b'** of the wall part **15'**.

(139) Having done that, the molten metal Q that reached the support part **155'** first becomes a wall, and the gas G generated from the region around the bolt boss **122b'** being pushed out toward the outlet port M4 side in the straight line Lm direction is inhibited, and is left behind in the mold MA. The gas G left behind in the molten metal Q in the transmission case **100** after casting appears as cavities (see FIG. 17).

(140) With the transmission case **1** according to the present invention, by connecting the rib **8** to the pin boss **124**, it becomes easier for the molten metal Q to be supplied during casting even in the region around the bolt boss **122b** of the wall part **15** which is the region in which the ribs are not formed. As a result, the difference in the supply rate of the molten metal Q to each part in the mold M becomes smaller, and as with the comparison example in FIG. 15 to FIG. 17, there is a reduction in the gas G generated from the molten metal Q becoming caught.

(141) Here, when the rib **8** is connected to the pin boss **124**, the gap between the rib **8** and the rib **7** adjacent in the circumferential direction around the rotation axis X becomes broader than the transmission case **100** according to the comparison example, but by providing the rib **9**, rigidity is ensured.

(142) As shown in FIG. 6, in the transmission case **1** according to the present embodiment, the straight line Lp1 along the longitudinal direction of the radial direction wall **91** of the rib **9** is made to pass above the center C1 of the bolt hole **120** of the bolt boss **122c**. As a result, the gap between the radial direction wall **91** and the short wall part **134** becomes broader. Having done that, it is possible to broaden the gap in the part configuring the radial direction wall **91** and the short wall part **134** in the mold M, so heat conveyed to the mold M from the molten metal Q can be dispersed more easily. As a result, there is a reduction in the occurrence of heat cracks with a local increase in the temperature of the mold M due to the heat of the molten metal Q.

(143) The straight line Lp2 along the longitudinal direction of the circumferential direction wall **92** of the rib **9** is made to pass below the center C2 of the bolt hole **120** of the bolt boss **121**. As a result, the gap between the circumferential direction wall **92** and the rib **7** becomes broader. Having done that, it is possible to broaden the gap in the part configuring the circumferential direction wall **92** and the rib **7** in the mold M, so heat conveyed to the mold M from the molten metal Q is dispersed more easily. As a result, there is a reduction in the occurrence of heat cracks in the mold M with a local increase of the temperature of the mold M due to the heat of the molten metal Q.

(144) Examples of the transmission case **1** according to a mode of the present invention are listed hereafter.

(145) (1, 2) The transmission case **1** (case) houses the transmission **61** of the drive force transmission device **6**.

(146) The transmission case **1** has the support part **155** (circular ring part) that forms the opening **150** through which the output shaft **63** (shaft) that transmits the drive force passes.

(147) The plurality of bolt bosses **121** are provided around the support part **155**.

(148) The bolt bosses **122a**, **122b**, **122c** are provided around the support part **155**.

(149) The bolt bosses **121**, **122a**, **122b**, **122c** are fastening boss parts.

(150) A bolt B (fastening member) is secured to each of the bolt bosses **121**, **122a**, **122b**, **122c**.

(151) Viewed from the rotation axis X direction (axial direction of the shaft), among the bolt bosses **121**, **122a**, **122b**, **122c** (plurality of fastening boss parts), the ribs **7** (first ribs) are formed between the bolt bosses **121** (portion of the fastening boss parts) and the support **155**.

(152) Among the bolt bosses **121**, **122a**, **122b**, **122c**, the bolt bosses **122a**, **122b**, **122c** are the plurality of other fastening boss parts to which the ribs **7** are not connected.

(153) The pin boss **124** is provided between the bolt bosses **122b**, **122c**.

(154) The pin boss **124** is the non-fastening boss part to which the bolt B is not secured.

(155) The pin boss **124** has the pin hole **124a** (recess) in which the pin P2 for positioning is inserted.

(156) The rib **8** (second rib) is formed between the pin boss **124** and the support **155**.

(157) By configuring in this way, it is possible to provide a transmission case **1** that reduces the occurrence of cavities and improves yield.

(158) In specific terms, in the transmission case **1**, by providing the rib **8** between the pin boss **124** and the support part **155**, the rib **8** is brought closer to the bolt boss **122b**. The bolt boss **122b** is the fastening boss part to which the ribs **7**, **8** are not connected.

(159) As a result, during casting, the supply of the molten metal Q to the bolt boss **122b'** side via the rib **8'** is promoted. Having done that, the difference in the supply rate of the molten metal Q to each part within the mold M becomes smaller, and it is possible to reduce the gas G generated from the molten metal Q becoming caught. Therefore, it is possible to provide the transmission case **1** that reduces the occurrence of cavities and improves yield.

(160) (3) Viewed from the rotation axis X direction, the pin boss **124** is provided between the lower end surface **13a** (one end face) of the wall part **13** in the radial direction of the rotation axis X and the support part **155**.

(161) Casting of the transmission case **1** is done by supplying the molten metal Q from the inlet ports M1 to M3 of the mold M.

(162) The inlet ports M1 to M3 of the molten metal Q are provided at the lower end surface **13a'** side of the wall part **13'** of the transmission case **1'** in the radial direction of the rotation axis X.

(163) By configuring in this way, during casting, the molten metal Q is supplied to the rib **8'** via the pin boss **124'** from the lower end surface **13a'** side on the wall part **13'**. As a result, the molten metal Q is more easily supplied also to the bolt boss **122b'** side which is the fastening boss part to which the ribs **7'**, **8'** are not connected. Having done that, the difference in the supply rate of the molten metal Q to each part in the mold M is smaller, and there is a reduction in the gas G generated from the molten metal Q becoming caught. Therefore, it is possible to provide a transmission case **1** that reduces the occurrence of cavities and improves yield.

(164) (4) The pin boss **124** is linked to the bolt boss **122c**.

(165) The bolt boss **122c** is one of the plurality of other fastening boss parts to which the ribs **7** are not connected.

(166) By configuring in this way, by having the transmission case **1** have the pin boss **124** and the bolt boss **122c** linked, it is possible to make the space in which the molten metal Q is pooled (pin boss **124'**, bolt boss **122c'**) larger in the mold M. As a result, during casting, there is a greater volume of the molten metal Q supplied via the rib **8'** from the pin boss **124'**, and the molten metal Q is more easily supplied also to the bolt boss **122b'** side. Having done that, it is possible to have the difference in supply rate of the molten metal Q to each part in the mold M be smaller, and to reduce the gas G generated from the molten metal Q becoming caught. Therefore, it is possible to

reduce the occurrence of cavities.

(167) In particular, it is possible to have the space in which the molten metal Q is pooled (pin boss **124'**, bolt boss **122c'**) be closer to the inlet port M3 in the mold M the closer the pin boss **124** is to the lower end surface **13a** of the wall part **13**. As a result, for example before the molten metal Q flows to the parallel wall **128'** side, it can be taken into the pin boss **124'** and the bolt boss **122c'**, so it is possible to increase the amount of the molten metal Q supplied to the rib **8'**.

(168) (5) There is the arc-shaped recess **122d** in the region in which the bolt boss **122c** is provided.

(169) The bolt boss **122c** is one of the plurality of other boss parts to which the ribs **7** are not connected.

(170) There is an arc-shaped recess **121a** in the region in which the bolt boss **121** is provided.

(171) Viewed from the rotation axis X direction, the radius r1 of the recess **122d** is larger than the radius r2 of the recess **121a**.

(172) By configuring in this way, in the bolt boss **122c** to which the rib **7** is not connected, it is possible to reduce the stress concentration when twisting occurs between it and the cover member **3**. As a result, it is possible to prevent cracking of the transmission case **1** due to stress concentration. Meanwhile, padding is reduced in the bolt bosses **121** to which the ribs **7** are connected, so it is possible to reduce the overall weight of the transmission case **1** from becoming heavy.

(173) (6) The rib **9** (third rib) is provided between the rib **7** and the rib **8** adjacent in the circumferential direction around the rotation axis X.

(174) The gap between ribs **7**, **8** adjacent in the circumferential direction around the rotation axis X is broader than the gap between ribs **7**, **7** adjacent in the circumferential direction around the rotation axis X.

(175) In light of that, by configuring as noted above, by providing the rib **9** around the bolt boss **122c** in which the ribs **7**, **8** are not provided, it is possible to ensure rigidity strength around the bolt boss **122c**.

(176) (7) The rib **9** cross section is a truss structure when viewed from the rotation axis X direction.

(177) By configuring in this way, it is possible to improve surface strength compared to simply providing one rib on the bolt boss **122c**.

(178) (8) The rib **9** has the radial direction wall **91** extending to the radial inner diameter side from the bolt boss **122c**, and the circumferential direction wall **92** that connects the radial direction wall **91** and the bolt boss **121** in the circumferential direction.

(179) The rib **9** constitutes a truss structure by connecting the three points of the intersection part K between the radial direction wall **91** and the circumferential direction wall **92**, the bolt boss **122c**, and the bolt boss **121**.

(180) By configuring in this way, it is possible to distribute the stress between the radial direction wall **91**, the circumferential direction wall **92**, and the parallel wall **128**.

(181) (9) In the radial direction wall **91**, the straight line Lp1 along the longitudinal direction passes above the center C1 of the bolt boss **122c** in the vertical line VL direction.

(182) In the circumferential direction wall **92**, the straight line Lp2 along the longitudinal direction passes below the center C2 of the bolt boss **121** in the vertical line VL direction.

(183) Specifically, viewed from the rotation axis X, the radial direction wall **91** and the circumferential direction **92** are provided at positions respectively offset in the direction approaching each other in the vertical line VL direction from the center C1 of the bolt boss **122c** and the center C2 of the bolt boss **121**.

(184) By configuring in this way, the radial direction wall **91** and the circumferential direction wall **92** are respectively disposed at positions away from the short wall part **134** and the ribs **7**.

(185) Having done that, it is possible to make the gap of the part corresponding to the radial direction wall **91** and the short wall part **134** in the mold M and the gap of the part corresponding to the circumferential direction wall **92** and the rib **7** broader, so it is easier to disperse heat conveyed

to the mold M from the molten metal Q. As a result, there is a reduction in the occurrence of heat cracks with a local increase in the mold M temperature due to the heat of the molten metal Q. (186) As shown in FIG. 7, the outer diameter side wall part **151** has the bottom surface **151a** orthogonal to the rotation axis X at the rib **7** side from the center C2 of the bolt boss **121** in the radial direction of the rotation axis X. The outer diameter side wall part **151** has the inclined plane **151b** that is inclined in the direction approaching the other end **10b** in the rotation axis X direction at the parallel wall **128** side from the center C2 of the bolt boss **121** in the radial direction of the rotation axis X. The inclined plane **151b** is inclined in the direction approaching the other end **10b** in the rotation axis X direction as it moves away from the bottom surface **151a** in the radial direction of the rotation axis X.

(187) In light of that, by configuring as described above, by offsetting the circumferential direction wall **92** in the radial direction of the rotation axis X from the center C2, it is possible to provide the circumferential direction wall **92** on the inclined plane **151b**. Having done that, it is possible to shorten the protrusion height from the outer diameter side wall part **151** in the rotation axis X direction by an amount ΔH compared to when providing the circumferential direction wall **92** on the bottom surface **151a**. As a result, there is a reduction of the amount of material required for forming the circumferential direction wall **92** while maintaining the truss structure function of the circumferential direction wall **92**.

(188) (10) The width T9 of the radial direction wall **91** and the circumferential direction wall **92** is thinner than the widths T7, T8 of the ribs **7**, **8**.

(189) By configuring in this way, it is possible to reduce the overall weight of the transmission case **1** from becoming heavier.

MODIFICATION EXAMPLE

(190) In the embodiments noted above, examples were shown of when the rib **8** is connected to the pin boss **124**, but the invention is not limited to this mode. The boss may be for something other than a pin. The rib **8** may also be disposed more to the bolt boss **122b** side than the bolt boss **122c** in the straight line HL1 direction. In the modification example hereafter, only the parts that differ from the present embodiment are explained.

(191) FIG. **18** is a drawing for explaining a transmission case **1A** according to the modification example.

(192) As shown in FIG. **18**, a projection part **14** is provided at a position adjacent to the bolt boss **122c** in the straight line HL1 direction on the short wall part **134** of the transmission case **1A**. The projection part **14** constitutes the non-fastening boss part to which the bolt B is not secured.

(193) The projection part **14** is provided at a position that is offset by the distance CL1 in the straight line HL1 direction to the bolt boss **122b** side from the bolt boss **122c**.

(194) A rib **8A** extends in the radial direction of the rotation axis X and connects the support part **155** and the projection part **14**. In the transmission case **1A** of the modification example, the rib **8A** is provided between the projection part **14** and the support part **155** and is brought closer to the bolt boss **122b**.

(195) Though an illustration is omitted, as a result, during casting, supplying of the molten metal Q through a rib **8A'** to the bolt boss **122b'** side is promoted, and it is possible to reduce the gas G generated from the molten metal Q becoming caught. Therefore, it is possible to provide the transmission case **1A** that reduces the occurrence of cavities and improves yield.

(196) The transmission case **1A** according to the modification example has the following configuration, for example.

(197) (1) The transmission case **1A** (case) houses the transmission **61** of the drive force transmission device **6**.

(198) The transmission case **1A** has the support part **155** (circular ring part) that forms the opening **150** through which the output shaft **63** (shaft) that transmits the drive force is inserted.

(199) The plurality of bolt bosses **121** are provided around the support part **155**.

- (200) Bolt bosses **122a**, **122b**, **122c** are provided around the support part **155**.
- (201) The bolt bosses **121**, **122a**, **122b**, **122c** are fastening boss parts.
- (202) The bolt B (fastening member) is secured to each of the bolt bosses **121**, **121a**, **122b**, **122c**.
- (203) Viewed from the rotation axis X direction (axial direction of the shaft), the ribs **7** (first ribs) are formed between the bolt bosses **121** (portion of the fastening boss parts) among the bolt bosses **121**, **122a**, **122b**, **122c** (the plurality of fastening boss parts) and the support part **155**.
- (204) Among the bolt bosses **121**, **122a**, **122b**, **122c**, the bolt bosses **122a**, **122b**, **122c** are the plurality of other fastening boss parts to which the ribs **7** are not connected.
- (205) The projection part **14** is provided between the bolt bosses **122b**, **122c**.
- (206) The projection part **14** is the non-fastening boss part to which the bolt B is not secured.
- (207) The rib **8A** (second rib) is formed between the projection part **14** and the support part **155**.
- (208) By configuring in this way, it is possible to provide the transmission case **1A** that reduces the occurrence of cavities and improves yield.
- (209) In the transmission case **1A** according to the modification example, the projection part **14** and the bolt boss **122c** are separated in the straight line HL1 direction, but they may also be linked to each other. As a result, in the mold M, it is possible to make the space in which the molten metal Q is pooled larger, and to increase the amount of the molten metal Q supplied to the rib **8A'**.
- (210) In the present embodiment, as an example of the case, we showed an example of the transmission case **1** mounted in a vehicle, but the invention is not limited to this mode. It is also possible to apply this to a case used for something other than a vehicle.
- (211) Above, embodiments of the present invention were explained, but the abovementioned embodiment merely shows one application example of the present invention, and the claims of the present invention are not limited to the specific configurations of the embodiments noted above. Suitable modifications are possible within the scope of technical concepts of the invention.

EXPLANATION OF CODES

(212) **1**, **1A**: Transmission case (case); **6**: Drive force transmission device; **7**: Rib (first rib); **8**, **8A**: Rib (second rib); **9**: Rib (third rib); **14**: Projection part (non-fastening boss part); **63**: Output shaft (shaft); **91**: Radial direction wall; **92**: Circumferential direction wall; **13a**: Lower end surface (one end face); **121**: Bolt boss (portion of fastening boss parts to which the first ribs are connected); **121a**: Recess; **122a**, **122b**, **122c**: Bolt boss (plurality of other fastening boss parts to which the first ribs are not connected); **122d**: Recess; **124**: Pin boss (non-fastening boss part); **124a**: Pin hole (recess); **13a**: Lower end surface (one end face); **150**: Opening; **155**: Support part (circular ring part); B: Bolt (fastening member); C1: Center (center of the plurality of other fastening boss parts to which the first ribs are not connected); Cs2: Center (center of the portion of fastening boss parts to which the first ribs are connected); K: Intersection part; M: Mold; P2: Pin; Q: Molten metal; and X: Rotation axis (shaft axis).

Claims

1. A case that houses a drive force transmission device, comprising: a circular ring part defining an opening through which a shaft that transmits a drive force is configured to be inserted; a plurality of fastening boss parts provided around the circular ring part, to which fastening members are to be secured; a plurality of first ribs respectively formed between a portion of the fastening boss parts and the circular ring part when viewed from an axial direction of the shaft; a non-fastening boss part provided between adjacent ones of the fastening boss parts to which the first ribs are not connected when viewed from the axial direction, a fastening member being not configured to be secured to the non-fastening boss part; and a second rib formed between the non-fastening boss part and the circular ring part, wherein the non-fastening boss part is a pin boss part defining a recess in which a pin for positioning is to be inserted, and the second rib extends along a line passing through the recess defined by the non-fastening boss part when viewed from the axial

direction of the shaft.

2. A case that houses a drive force transmission device, comprising: a circular ring part defining an opening through which a shaft that transmits a drive force is configured to be inserted; a plurality of fastening boss parts provided around the circular ring part, to which fastening members are to be secured; a plurality of first ribs respectively formed between a portion of the fastening boss parts and the circular ring part when viewed from an axial direction of the shaft; a non-fastening boss part provided between adjacent ones of the fastening boss parts to which the first ribs are not connected when viewed from the axial direction, a fastening member being not configured to be secured to the non-fastening boss part; and a second rib formed between the non-fastening boss part and the circular ring part, the second rib extending along a line passing through a hole or a recess defined by the non-fastening boss part when viewed from the axial direction of the shaft, wherein when viewed from the axial direction, the non-fastening boss part is provided between one end face of the case in a radial direction of the shaft and the circular ring part, and casting of the case is done by supplying molten metal from a side of the one end face in the radial direction.

3. The case according to claim 2, wherein the non-fastening boss part is linked to one of the fastening boss parts to which the first ribs are not connected.

4. The case according to claim 3, further comprising an arc-shaped recess portion provided in each of a region in which the fastening boss parts to which the first ribs are not connected are provided, and a region in which the portion of the fastening boss parts to which the first ribs are connected are provided, and when viewed from the axial direction, a radius of the recess portion of the region in which the fastening boss parts to which the first ribs are not connected are provided is greater than a radius of the recess portion of the region in which the portion of the fastening boss parts to which the first ribs are connected are provided.

5. The case according to claim 4, further comprising a third rib provided between one of the first ribs and the second rib adjacent in a circumferential direction around an axis of the shaft.

6. The case according to claim 5, wherein a cross section of the third rib is a truss structure when viewed from the axial direction.

7. The case according to claim 6, wherein the third rib includes a radial direction wall extending to a radial inner diameter side of the shaft from one of the fastening boss parts to which the first ribs are not connected, and a circumferential direction wall in which the radial direction wall and one of the portion of the fastening boss parts to which the first ribs are connected are connected in the circumferential direction, wherein the third rib constitutes the truss structure by connecting three points of an intersection part of the radial direction wall and the circumferential direction wall, one of the fastening boss parts to which the first ribs are not connected, and one of the portion of the fastening boss parts to which the first ribs are connected.

8. The case according to claim 7, wherein when viewed from the axial direction, the radial direction wall and the circumferential direction wall are provided at positions each offset in a direction approaching each other from a center of one of the fastening boss parts to which the first ribs are not connected and a center of one of the portion of the fastening boss parts to which the first ribs are connected.

9. The case according to claim 8, wherein the radial direction wall and the circumferential direction wall are thinner than the first ribs and the second rib.

10. The case according to claim 1, wherein when viewed from the axial direction, the non-fastening boss part is provided between one end face of the case in a radial direction of the shaft and the circular ring part, and casting of the case is done by supplying molten metal from a side of the one end face in the radial direction.

11. The case according to claim 10, wherein the non-fastening boss part is linked to one of the fastening boss parts to which the first ribs are not connected.

12. A case that houses a drive force transmission device, comprising: a circular ring part defining an opening through which a shaft that transmits a drive force is to be inserted; a plurality of

fastening boss parts provided around the circular ring part, to which fastening members are to be secured; a plurality of first ribs respectively connected to outer circumferences of a portion of the fastening boss parts and the circular ring part when viewed from an axial direction of the shaft; a non-fastening boss part provided between adjacent ones of the fastening boss parts to which the first ribs are not connected when viewed from the axial direction, a fastening member being not configured to be secured to the non-fastening boss part; and a second rib formed between the non-fastening boss part and the circular ring part, the second rib extending along a line passing through a hole or a recess defined by the non-fastening boss part when viewed from the axial direction of the shaft.

13. The case according to claim 12, wherein the non-fastening boss part is a pin boss part having the recess in which a pin for positioning is to be inserted.

14. The case according to claim 12, wherein when viewed from the axial direction, the non-fastening boss part is provided between one end face of the case in a radial direction of the shaft and the circular ring part, and casting of the case is done by supplying molten metal from a side of the one end face in the radial direction.

15. The case according to claim 14, wherein the non-fastening boss part is linked to one of the fastening boss parts to which the first ribs are not connected.

16. The case according to claim 15, further comprising an arc-shaped recess portion provided in each of a region in which the fastening boss parts to which the first ribs are not connected are provided, and a region in which the portion of the fastening boss parts to which the first ribs are connected are provided, and when viewed from the axial direction, a radius of the recess portion of the region in which the fastening boss parts to which the first ribs are not connected are provided is greater than a radius of the recess portion of the region in which the portion of the fastening boss parts to which the first ribs are connected are provided.

17. The case according to claim 16, further comprising a third rib provided between one of the first ribs and the second rib adjacent in a circumferential direction around an axis of the shaft.

18. The case according to claim 17, wherein a cross section of the third rib is a truss structure when viewed from the axial direction.

19. The case according to claim 18, wherein the third rib includes a radial direction wall extending to a radial inner diameter side of the shaft from one of the fastening boss parts to which the first ribs are not connected, and a circumferential direction wall in which the radial direction wall and one of the portion of the fastening boss parts to which the first ribs are connected are connected in the circumferential direction, wherein the third rib constitutes the truss structure by connecting three points of an intersection part of the radial direction wall and the circumferential direction wall, one of the fastening boss parts to which the first ribs are not connected, and one of the portion of the fastening boss parts to which the first ribs are connected.

20. The case according to claim 19, wherein when viewed from the axial direction, the radial direction wall and the circumferential direction wall are provided at positions each offset in a direction approaching each other from a center of one of the fastening boss parts to which the first ribs are not connected and a center of one of the portion of the fastening boss parts to which the first ribs are connected.

21. The case according to claim 20, wherein the radial direction wall and the circumferential direction wall are thinner than the first ribs and the second rib.
